

Technicians FAQ – Smart Irrigation System

This FAQ is for technicians and operators. It explains the system in **simple, non-technical terms**.

1. What does the system do?

- Monitors soil moisture, temperature, humidity, rainfall, and water tank level.
 - Automatically decides **when and how much to water the crops**.
 - Sends data to a dashboard for monitoring and historical logs.
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2. How do I know the system is working?

- Check the **web dashboard** or **mobile app** for real-time sensor readings.
 - Pump status will be shown: **ON**, **OFF**, or **DELAY**.
 - Alerts or warnings will appear if any sensor fails.
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3. What if a sensor stops working?

- The system uses the **last valid reading** if a sensor fails.
 - A faulty sensor must be **replaced or recalibrated**.
 - Check the ESP32 gateway and wiring first.
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4. Can I manually control irrigation?

- Yes, through the **dashboard** or **mobile app**.
 - The rule engine will temporarily pause when manual control is used.
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5. What do the ET and rule engine mean?

- **ET (Evapotranspiration)**: measures how much water plants need.
 - **Rule Engine**: decides **whether to water or not**, based on soil moisture, ET, and weather forecast.
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6. What to do if the dashboard shows wrong readings?

- Refresh the dashboard.
 - Check sensor connections.
 - Verify the backend API is running: `http://<server-ip>/api/ping`.
 - Check logs in Docker containers: `docker logs -f irrigation-backend`.
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7. How do I update the system?

- Pull the latest code from GitHub.

- Restart services in this order:
 1. Database (`docker-compose up -d db`)
 2. Backend (`docker-compose up -d backend`)
 3. Swagger UI (`docker-compose up -d swagger`)
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8. Who to contact if something is broken?

- If hardware (sensors, pump) → contact **IoT team**.
 - If software (backend, dashboard) → contact **development team**.
 - Always **document the problem** before escalating.
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9. Daily Routine Checks

- Verify pump status and logs.
 - Ensure all sensors are reporting correctly.
 - Check water tank levels.
 - Review system warnings and alerts.
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10. Emergency Actions

- Turn off the pump manually if water overflows.
- Reboot the ESP32 gateway if sensors are unresponsive.
- Restart backend containers if the dashboard or rule engine stops responding.

11. How do I adjust soil moisture thresholds?

- Thresholds are defined in **rule-engine/engine.py** as constants: **LOWER_LIMIT = 30** and **UPPER_LIMIT = 70**. Edit these values and restart the backend.